Total No. of Questions : 8]

SEAT No. :

PE2307

[Total No. of Pages : 2

[6584]-216

B.E. (Mechanical Engineering)

ELECTRICAL AND HYBRID VEHICLE

(2019 Pattern) (Semester - VIII) (402051 E) (Elective - VI)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7or Q.8***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Use of electronic pocket calculator is allowed.***
5) ***Assume suitable data, if necessary.***

Q1) a) Explain with neat sketch construction and working of brushless DC motor and also state its advantages and disadvantages. **[9]**

b) Determine the rating of motor required for following data: Gross curb weight (GCW) = 100 Kg , load on vehicle =70 kg, Maximum speed = 50 kmph., Road friction = 0.01, Coefficient of drag =0.9, Density of air = 1.2 kg/m^3, Frontal area of vehicle = 0.6 m^2 ,Wheel radius= 0.28 m. Suggest the motor rating to cover the distance of 90 kms per charge. **[9]**

OR

Q2) a) Explain with neat sketch construction and working of Li-ion battery used for EV. **[9]**

b) In e-auto, the battery rating is 20Ahs at 24 volts. It has maximum speed of 25 kmph and motor has power of 250 Watts. How much distance it will cover. Assume suitable data.

Case 1 Using lead acid battery (Assume power available 50%)

Case 2 Using Li-ion battery (Assume power available 95%)

How much range will increase? **[9]**

Q3) a) Compare between mechanical differential and electric differential. **[8]**

b) Explain the concept of fuel efficiency analysis and its significant features for electric vehicle? **[9]**

OR

P.T.O.

Q4) a) Electric vehicle has the following attributes :drag coefficient of 0.20 with vehicle frontal cross sectional area $2.5m^2$, available battery energy of 25kWh.Assuming density of air 1.2 kg/m^3.At particular moment with a vehicle speed of 100km/hour, calculate aerodynamic drag force, power and distance covered for driving condition of i) calm condition with no wind ii) windy conditions with a 12 km/h headwind. **[9]**

b) Write a note on braking system in EV and their types **[8]**

Q5) a) What is retrofitting and explain problems associated with retrofitting of two wheeler. **[9]**

b) Explain with neat sketch construction and working of rear end suspension system in vehicles. **[9]**

OR

Q6) a) What is homologation? What are the national/international testing/ regulation/licensing/approval organizations and agencies? **[9]**

b) Write a note on crash testing and testing of tracks. **[9]**

Q7) a) Explain with neat sketch charging architecture. **[8]**

b) Explain following: grid voltages, frequencies and wiring, real power, apparent power and power factor. **[9]**

OR

Q8) a) Describe and illustrate a typical structure of battery management systems (BMS) along with its significant functions. **[9]**

b) Describe and illustrate hazard/safety management of batteries during/after operations. **[8]**

✪ ✪ ✪